TAWHIDUL ISLAM

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SUMMARY

Full-Stack Developer with hands-on experience building web/mobile applications, event-driven backends, and cloud architectures. Passionate about AI integration, backend performance, and intuitive user experiences. Proven ability to design and ship end-to-end projects using modern frameworks, databases, and cloud services.

TECHNICAL SKILLS

Languages:	C++, Java, Python, Dart, JavaScript (ES6+), TypeScript, SQL (PostgreSQL, MySQL, SQLite)
Web & Mobile:	Flutter, React, Vite, Node.js, Express, Flask, FastAPI, HTML5, CSS3, REST APIs
Databases & AI:	PostgreSQL, Supabase, pgvector, Redis, Apache Kafka, PyTorch, OpenCV, MediaPipe
Tools & Platforms:	AWS (S3), GCP, Firebase, Docker, Git, GitHub/GitLab, OpenAI API, LibreTranslate

WORK EXPERIENCE

Full Stack Engineer Intern | Zetheta Algorithms Private Limited Aug 2026 – Oct 2026

- Engineered a production-grade, 8-step loan application form featuring real-time schema validation, conditional rendering, secure document uploads with client-side compression, and e-signature capture.
- Implemented a custom API integration framework connecting enterprise SAP S/4HANA systems to a financial analytics platform using secure OAuth 2.0 protocols and robust ETL transformation logic.
- Architected a multi-region active-active disaster recovery framework on AWS using Aurora PostgreSQL global replication and Kafka streaming to ensure 99.99% uptime.
- Developed an event-driven notification backend with Apache Kafka and Redis, routing 25+ real-time event types across 5 channels with TRAI DND compliance, rate limiting, and retries.

PROJECTS

Translify – Real-Time Multimodal Translation Mobile Platform
Stack: Flutter, Dart, Python, FastAPI, OpenCV, Pytesseract, LibreTranslate, Docker, PostgreSQL

- Architected a privacy-focused mobile translation application supporting camera OCR scanning and live conversational speech-to-text.
- Engineered a FastAPI microservices backend integrating OpenCV, Pytesseract, and self-hosted LibreTranslate via Docker.

OpenBook – Flutter eBook Reader & Audio Library
Stack: Flutter, Dart, Iridium EPUB Engine, Gutendex API, TTS, SQLite

- Built a mobile eBook reader with live search debouncing, custom shelves, offline SQLite caching, and text-to-speech audio controls.
- Integrated the Iridium EPUB engine and Gutendex API to parse public-domain literature for seamless reader navigation.

Journey – AI-Assisted Mobile Fitness Tracker
Stack: Flutter, Dart, Flask (Python), Supabase (PostgreSQL), OpenAI GPT API

- Developed a cross-platform Flutter app for workout safety and form correction, leveraging OpenAI GPT API for real-time guidance.
- Architected a lightweight Flask API backend and Supabase database to stream real-time progression metrics.

IntelliDoc – AI-Powered Document Management System
Stack: React, Vite, FastAPI, Python, PostgreSQL, pgvector, SQLAlchemy, Alembic, AWS S3, PyTorch

- Engineered a full-stack document system with semantic file search powered by BAAI/bge-m3 embeddings and PyTorch.
- Implemented vector search with PostgreSQL + pgvector, FastAPI microservices, and a responsive React interface.

EDUCATION

B.A. in Computer Science – Hunter College, CUNY Sept 2023 – May 2026
A.S. in Computer Science – LaGuardia Community College, CUNY Sept 2021 – Aug 2023